

EPC2361

Double-pulse simulation datasheet

Setup, waveforms and switching-energy comparisons

16 September 2026

Revision 1

Study conditions: 75 V DC supply, fixed 100 °C model temperature, +5 / -4 V gate-drive rails. Five example configurations; nominal 0, 10, 20, 30, 40 and 50 A.

Purpose and interpretation

This report characterizes the supplied EPC2361 SPICE model in a low-side double-pulse test (DPT). It retains the existing GUI and library. It is a reproducible simulation comparison, not measured hardware data, a guaranteed device specification, or a qualified 75 V / 50 A design.

Example configurations

The resistances below are external gate resistors. The driver adds 0.7 Ω pull-up and 0.4 Ω pull-down resistance; the EPC library also contains its own device model.

ID	Comparison	$R_{g,on}$ (Ω)	$R_{g,off}$ (Ω)	L_{layout} (nH)
A	Reference	2	0.5	1
B	Higher layout inductance	2	0.5	5
C	Higher on/off resistances	5	2	1
D	Isolate on-resistance change	5	0.5	1
E	Isolate off-resistance change	2	2.5	1

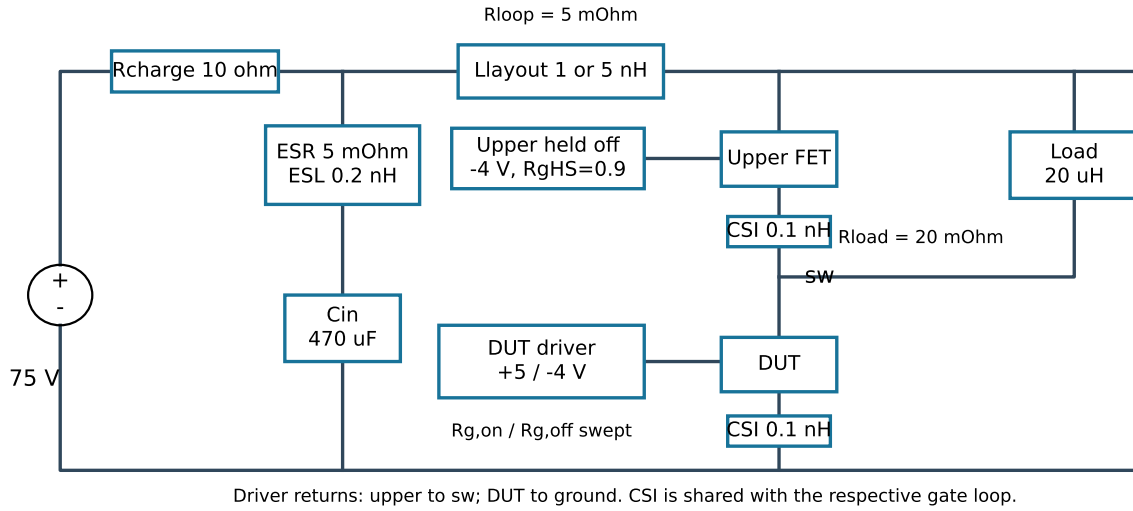
Limits that affect these results

The EPC2361 datasheet specifies 100 V continuous drain-source voltage and 120 V repetitive transient voltage only at a duty factor of at most 1%. Gate-source limits are +6 / -4 V. Thus the requested -4 V off rail leaves no negative-undershoot margin. A SPICE model can still return numbers beyond these limits; those numbers do not establish device survival or accuracy outside the ratings [1]. Of 30 completed runs, 18 exceed 100 V, 6 exceed 120 V on at least one device, and 30 cross below -4 V gate-source voltage. See the stress graphs and per-run CSV for details.

What is included

A setup explanation, full switching waveforms, event zooms, signed E_{on}/E_{off} versus measured current, both-device voltage stress, gate undershoot, numerical checks and tabulated data. Each event integral uses the explicitly specified window; it must not be substituted uncritically for datasheet loss values.

1. DPT setup and measurement definition



The load inductor connects DC+ to the switching node. The lower EPC2361 is the DUT. Its first pulse builds load current, the off interval commutates current through the upper device in reverse conduction, and the second pulse measures loaded turn-on. The upper gate stays at -4 V relative to its driver return; no synchronous upper pulse is applied.

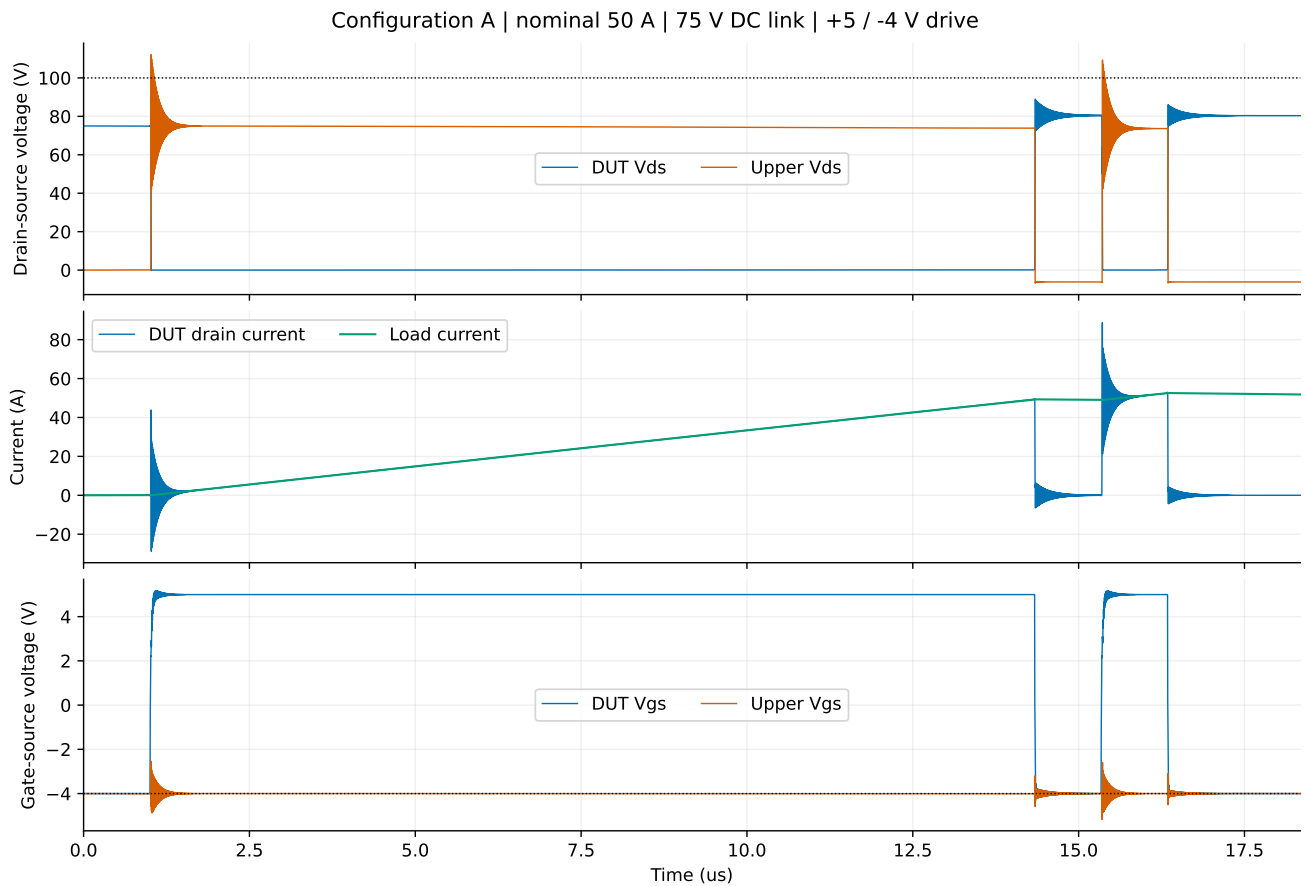
Quantity	Setting	Quantity	Setting
Supply / temperature	75 V / 100 °C	Load inductance / DCR	20 μ H / 20 m Ω
DC-link capacitance	470 μ F	Capacitor ESR / ESL	5 m Ω / 0.2 nH
Charging resistance	10 Ω	Layout resistance	5 m Ω
CSI per device	0.1 nH	Upper off-state gate R	0.9 Ω
Initial delay / off gap	1 / 1 μ s	Second pulse / tail	1 / 2 μ s
Command rise / fall	2 / 2 ns	Nominal maximum step	0.5 ns

The capacitor charges through the finite supply resistance during the DC operating-point solution. The first pulse is estimated as $T_1 = I_{target}L/V_{DC}$. Parasitic drops, finite edges and reverse conduction mean that actual current differs from the nominal setting. Each graph uses the measured current immediately before its corresponding event. **Energy definition.** E_{off} is from the first turn-off; E_{on} is from the second turn-on. With t_e the start of the corresponding command edge:

$$E_{event} = 10^6 \int_{t_e - 20 \text{ ns}}^{t_e + 100 \text{ ns}} v_{DS,DUT}(t) i_{D,DUT}(t) dt \quad [\mu\text{J}].$$

Current is positive into the DUT drain. These signed terminal-energy integrals include capacitive exchange, ringing and a small conduction contribution. No absolute-value rectification or forced monotonic fitting is used. Gate-driver energy and upper-device energy are excluded. **Nominal 0 A.** A separate no-load companion uses a 1 MH load approximation. It is shown with isolated open-square markers. Without load current, the DUT may not recover the blocking voltage before pulse 2; this is not a zero-current 75 V hard-switching loss measurement.

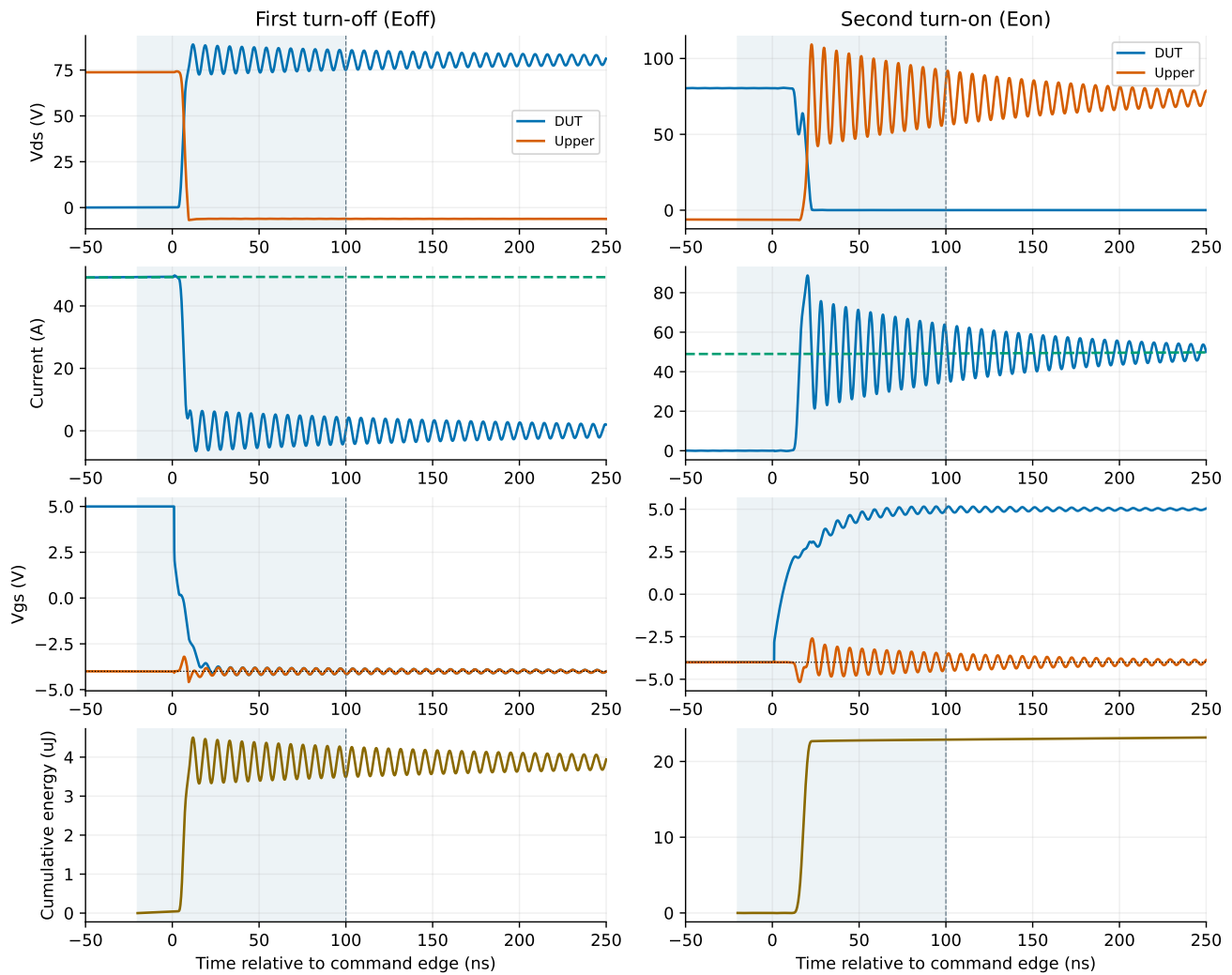
2. Full switching waveforms



Reference A, nominal 50 A: measured first-turn-off current 49.283 A and second-turn-on current 48.953 A. The upper-device waveform is its drain-to-source voltage, not its drain voltage relative to ground.

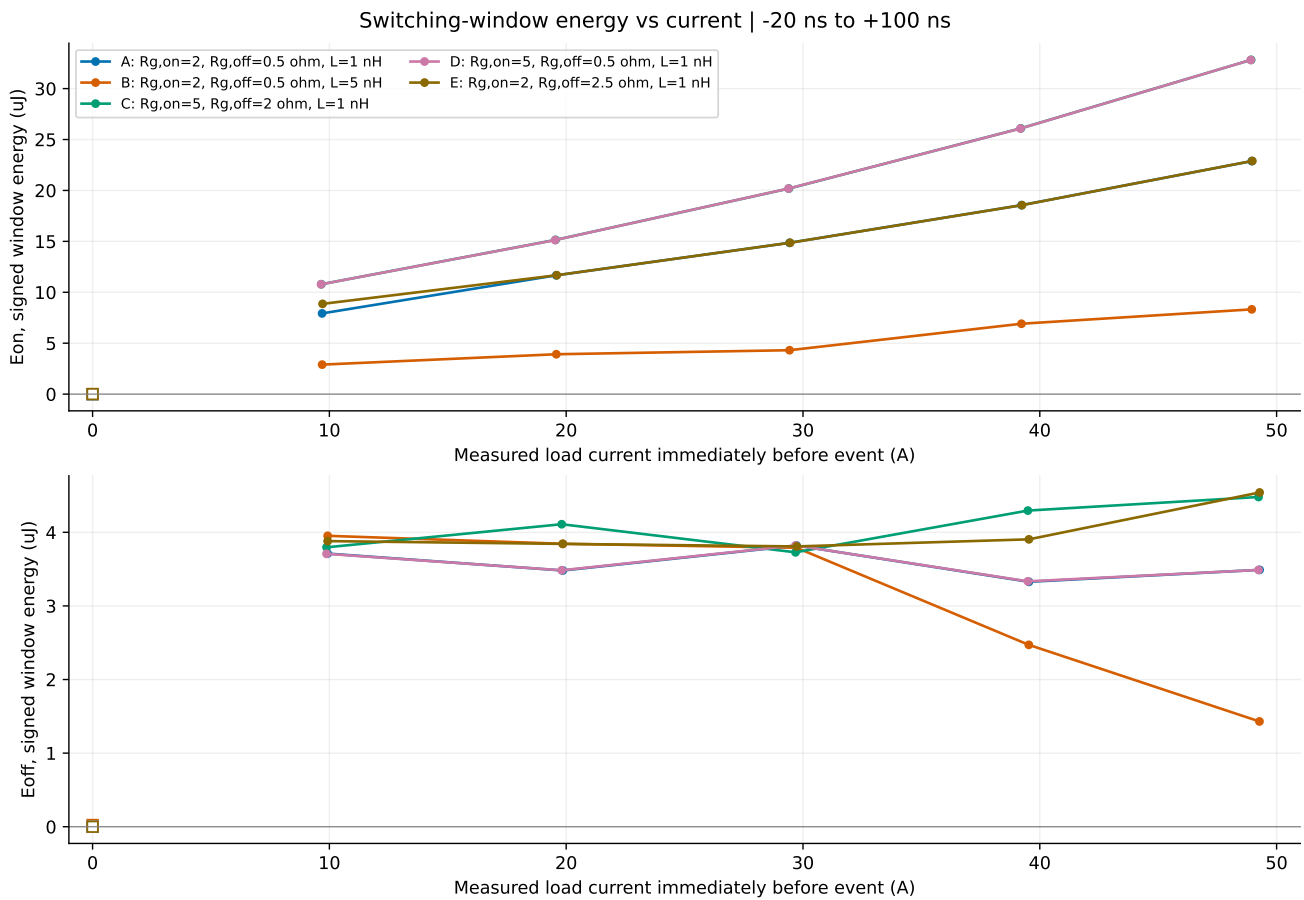
The negative gate rail applies to both off-state gate drivers. The upper device must remain off while the lower device is on. Adding an upper on-command at the instant of low-side turn-on would short the DC link; synchronous freewheeling is a different switching sequence and requires dead time.

3. Switching transitions and energy accumulation



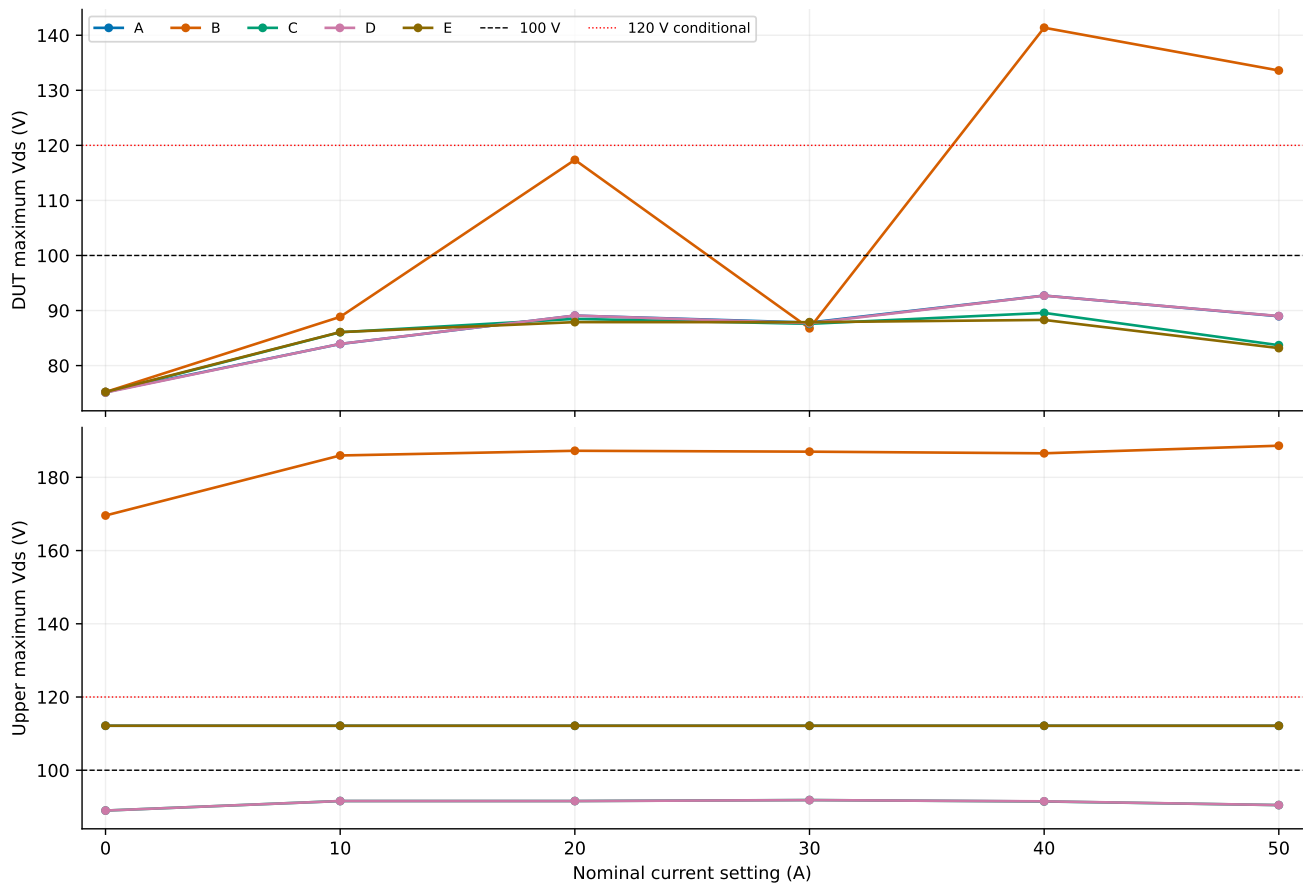
Shading identifies the integration window. The cumulative-energy trace continues to +250 ns to expose any remaining oscillatory energy exchange; the reported value is sampled at +100 ns. A different endpoint can change the signed energy, even when peak voltage is nearly unchanged.

4. Switching energy versus current



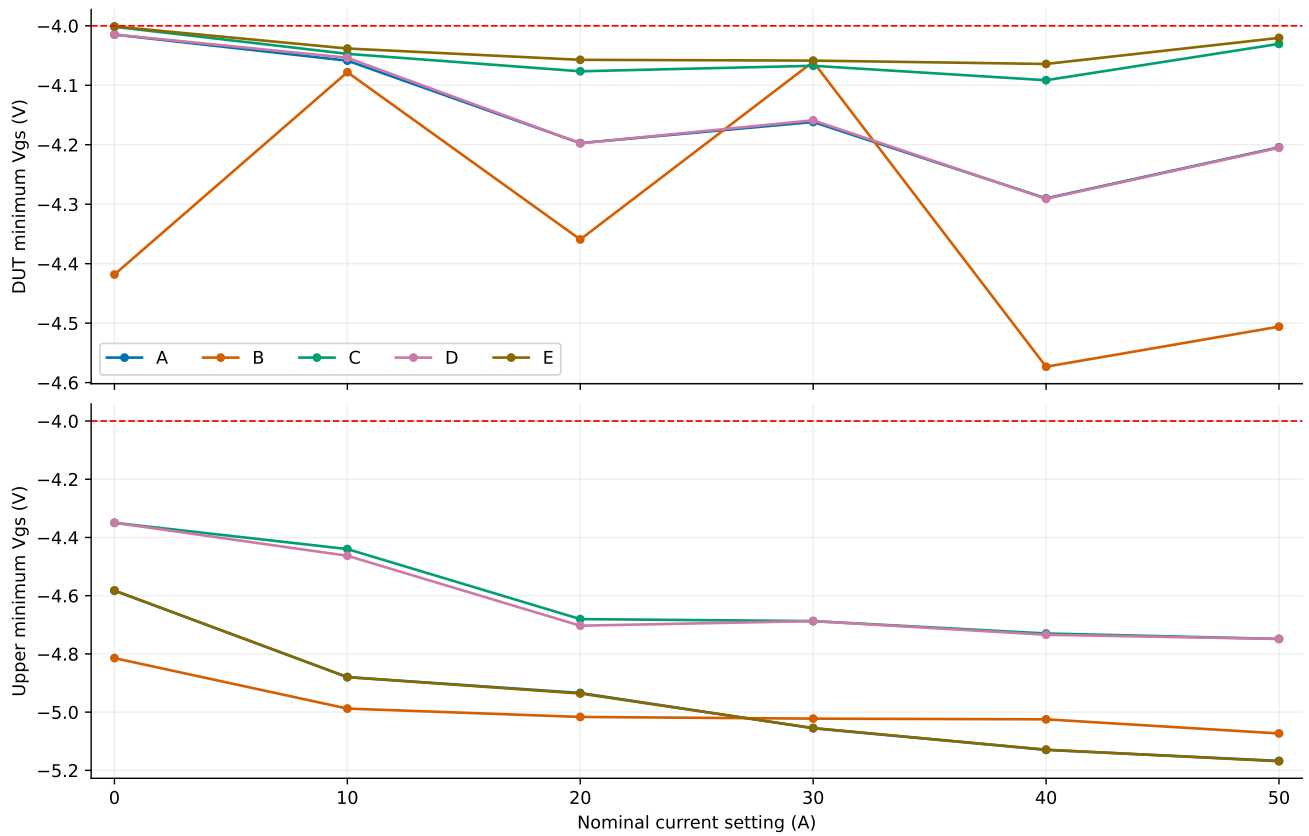
Lines connect the 10–50 A nominal settings, plotted at actual pre-event current. Open squares near 0 A are the separate no-load companion runs. Resistance values are in ohms. The curves are raw signed-window simulation results, including runs that exceed the device ratings; they are not qualified loss specifications. Increasing gate resistance changes transition duration as well as voltage overshoot. Increasing layout inductance changes resonant energy exchange. A smaller signed energy in a high-overshoot case does not establish improved efficiency.

5. Drain-source voltage stress



Peak values cover the entire DPT sequence, including first turn-on and the observation tail. The 120 V line is shown for context only; its datasheet duty-factor condition has not been qualified here. A curve below 100 V in this study alone does not establish worst-case voltage margin for a physical half-bridge. The load-current targets, pulse widths and gate resistors do not define actual PCB inductance. The selected 1–5 nH values and 0.2 nH capacitor ESL are circuit assumptions that need layout extraction or measurement.

6. Gate stress with -4 V off drive



The horizontal line is the negative gate-source rating, not a recommended design target. Actual Vgs includes common-source inductance and capacitive coupling; a -4 V source setting does not clamp device-terminal Vgs to -4 V. The positive peaks and all per-run limit flags are also included in the CSV. EPC identifies 0 V as the typical off drive and notes that negative bias increases reverse drain-source voltage. The requested -4 V case is retained here as a simulation experiment, not recommended as a gate-drive design [1].

7. Numerical checks and interpretation

Gear integration is used consistently for the sweep, with maximum timestep 0.5 ns, relative tolerance 10^{-4} , absolute current tolerance 10^{-9} A, voltage tolerance 10^{-6} V, and waveform compression disabled. Binary waveforms are stored in double precision. Energies are recomputed with trapezoidal integration after interpolating the exact window endpoints.

Quantity	0.5 ns	0.25 ns	Change (%)
E_{on} (μJ)	22.8813	22.8833	+0.01
E_{off} (μJ)	3.4899	3.4884	-0.04
DUT peak (V)	88.9629	88.9616	-0.00
Upper peak (V)	112.1454	112.1421	-0.00

Window sensitivity at 50 A nominal, configuration A:

E_{on} : 22.8813 μJ at +100 ns versus 23.0701 μJ at +200 ns.

E_{off} : 3.4899 μJ at +100 ns versus 3.8127 μJ at +200 ns.

Agreement in one case is not convergence of every configuration. Continuing ringing and window dependence limit how confidently these values can be interpreted as dissipated switching loss. No semiconductor failure, dynamic self-heating, load-inductor saturation, model variation or full repetitive converter operation is established by this study.

Relating the study to the GUI

The existing GUI is unchanged. It can still load saved settings and run individual DPTs. This report uses a separate sweep script, with its numerical options recorded in each saved netlist and schematic. The GUI's frequency-based power estimate is $P_{sw} = (E_{on} + E_{off})f_{sw}$ after converting microjoules to joules; it is an event-energy extrapolation, not a full converter loss simulation. For conduction, the GUI uses independently entered ON-interval RMS current, ON fractions and operating-temperature $R_{ds(on)}$. That estimate is distinct from these signed switching integrals and from reverse conduction during this DPT.

Reproduction

Run `python datasheet_sweep.py` followed by `python build_datasheet.py`. Successful case results are cached; use `--force` to rerun. Each run retains its JSON settings, ASC schematic, CIR netlist, LTspice log/raw data, reduced NPZ waveform and result JSON. Summary data are in `results.csv`. SHA-256 hashes of the model and GUI are recorded in `provenance.json`.

8. Numerical data: configurations A–C

A: $R_{g,on} = 2\ \Omega$, $R_{g,off} = 0.5\ \Omega$, $L_{layout} = 1\ \text{nH}$

Target A	I_{on} A	I_{off} A	E_{on} μJ	E_{off} μJ	DUT peak V	Upper peak V
0	0.000	0.000	0.000	-0.000	75.203	112.153
10	9.693	9.927	7.926	3.711	83.927	112.145
20	19.581	19.856	11.667	3.481	89.079	112.145
30	29.433	29.730	14.858	3.820	87.816	112.145
40	39.227	39.541	18.554	3.328	92.716	112.145
50	48.953	49.283	22.881	3.490	88.963	112.145

B: $R_{g,on} = 2\ \Omega$, $R_{g,off} = 0.5\ \Omega$, $L_{layout} = 5\ \text{nH}$

Target A	I_{on} A	I_{off} A	E_{on} μJ	E_{off} μJ	DUT peak V	Upper peak V
0	0.000	0.000	0.014	0.024	75.197	169.579
10	9.694	9.928	2.897	3.952	88.850	185.974
20	19.582	19.855	3.916	3.844	117.366	187.258
30	29.433	29.727	4.312	3.790	86.772	187.014
40	39.226	39.536	6.912	2.471	141.370	186.569
50	48.950	49.276	8.323	1.430	133.601	188.641

C: $R_{g,on} = 5\ \Omega$, $R_{g,off} = 2\ \Omega$, $L_{layout} = 1\ \text{nH}$

Target A	I_{on} A	I_{off} A	E_{on} μJ	E_{off} μJ	DUT peak V	Upper peak V
0	0.000	0.000	0.000	-0.000	75.101	88.985
10	9.662	9.881	10.791	3.796	86.055	91.587
20	19.551	19.811	15.144	4.110	88.485	91.587
30	29.404	29.685	20.184	3.728	87.582	91.853
40	39.198	39.496	26.097	4.295	89.574	91.470
50	48.925	49.238	32.822	4.480	83.704	90.479

9. Numerical data: configurations D–E

D: $R_{g,on} = 5 \, \Omega$, $R_{g,off} = 0.5 \, \Omega$, $L_{layout} = 1 \, \text{nH}$

Target A	I_{on} A	I_{off} A	E_{on} μJ	E_{off} μJ	DUT peak V	Upper peak V
0	0.000	0.000	0.000	-0.000	75.101	88.983
10	9.647	9.881	10.782	3.708	83.939	91.590
20	19.536	19.811	15.128	3.484	89.088	91.582
30	29.388	29.685	20.185	3.822	87.699	91.842
40	39.182	39.496	26.078	3.334	92.705	91.503
50	48.909	49.238	32.811	3.489	88.999	90.485

E: $R_{g,on} = 2 \, \Omega$, $R_{g,off} = 2.5 \, \Omega$, $L_{layout} = 1 \, \text{nH}$

Target A	I_{on} A	I_{off} A	E_{on} μJ	E_{off} μJ	DUT peak V	Upper peak V
0	0.000	0.000	0.000	-0.000	75.202	112.153
10	9.712	9.927	8.862	3.881	86.082	112.146
20	19.602	19.856	11.675	3.843	87.894	112.146
30	29.454	29.730	14.862	3.808	87.886	112.146
40	39.249	39.541	18.556	3.905	88.295	112.146
50	48.975	49.283	22.895	4.541	83.177	112.146

Source documents

- [1] Efficient Power Conversion, *EPC2361 eGaN FET datasheet*, revision July 20, 2026, pp. 1–3. https://epc-co.com/epc/Portals/0/epc/documents/datasheets/EPC2361_datasheet.pdf. Accessed September 16, 2026. The supplied library version is identified by its hash, not assumed to match this datasheet revision.
- [2] Wolfspeed, *Power Module SPICE Models User Guide*, PRD-07913 Rev. 2, December 2023, pp. 7–8, supplied PDF. Used for the clamped-inductive DPT arrangement and external-parasitic interpretation; no Wolfspeed SiC model was substituted.
- [3] D. Levett, Z. Zheng and T. Frank, Infineon, *Double Pulse Testing: The How, What and Why*, Bodo’s Power Systems, April 2020, supplied PDF, pp. 1 and 4. Used for the pulse sequence and measurements of current, voltage and switching behavior.